

Problem 2.12

Suppose that an accumulation function $a(t)$ is differentiable and satisfies the property

$$a(s+t) = a(s) + a(t) - a(0)$$

for all non-negative real numbers s and t .

1. Using the definition of derivative as a limit of a difference quotient, show that $a'(t) = a'(0)$.
2. Show that $a(t) = 1 + it$ where $i = a(1) - a(0) = a(1) - 1$.

Solution

(a) Show that $a'(t) = a'(0)$

By definition,

$$a'(0) = \lim_{h \rightarrow 0} \frac{a(0+h) - a(0)}{h}.$$

Since the given property implies

$$a(0+h) = a(0) + a(h) - a(0) = a(h),$$

we get

$$a'(0) = \lim_{h \rightarrow 0} \frac{a(h) - a(0)}{h}.$$

Now compute $a'(t)$:

$$a'(t) = \lim_{h \rightarrow 0} \frac{a(t+h) - a(t)}{h}.$$

Using the functional relation with $s = t$,

$$a(t+h) = a(t) + a(h) - a(0).$$

Substituting this into the difference quotient,

$$a'(t) = \lim_{h \rightarrow 0} \frac{a(t) + a(h) - a(0) - a(t)}{h}.$$

Thus,

$$a'(t) = \lim_{h \rightarrow 0} \frac{a(h) - a(0)}{h}.$$

But this is exactly the expression obtained for $a'(0)$, so

$$\boxed{a'(t) = a'(0)}.$$

(b) Show that $a(t) = 1 + it$ where $i = a(1) - a(0) = a(1) - 1$

Since $a'(t) = a'(0)$ for all t , the derivative is constant. Therefore $a(t)$ must be a linear function of the form

$$a(t) = mt + b.$$

Because $a(0) = 1$ for an accumulation function, we have

$$b = 1.$$

So

$$a(t) = mt + 1.$$

Now evaluate at $t = 1$:

$$a(1) = m + 1.$$

Hence,

$$m = a(1) - 1.$$

If we define

$$i = a(1) - a(0) = a(1) - 1,$$

then $m = i$, and therefore

$$\boxed{a(t) = 1 + it}.$$

The graph is therefore a straight line passing through $(0, 1)$ with slope i :

